Hvem er grafen?

Et spill for to, som «Gjett hvem». Begge har et hemmelig kort med en rett linje. Ved å stille ja/nei-spørsmål om stigningstall, konstantledd og skjæringspunkter prøver de å finne motstanderens graf. Varighet: 15–20 minutter.

Forberedelser

1. Skriv ut side 2 til hver elev, og side 3–4 (hemmelige kort) til hvert par. Klipp ut kortene.

Slik spiller dere

1. Begge trekker et hemmelig kort uten å vise det.
2. Bytt på å stille ett ja/nei-spørsmål om gangen, for eksempel: «Er stigningstallet negativt?», «Skjærer grafen y-aksen over 0?», «Går grafen gjennom (0, 0)?»
3. Kryss ut grafene på brettet som ikke passer med svaret.
4. Den som først gjetter riktig graf, og kan si funksjonsuttrykket, vinner.

Ikke lov

Det er ikke lov å spørre om bokstaven eller hele funksjonsuttrykket direkte. Spørsmålene skal handle om egenskaper ved grafen.

Til læreren: alle grafene

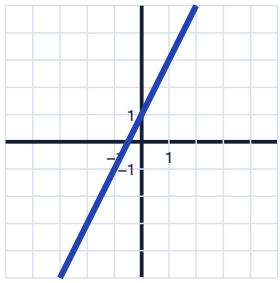
A $y = 2x + 1$	**B** $y = 2x - 2$	**C** $y = x + 3$	**D** $y = x$
E $y = x - 2$	**F** $y = -x + 3$	**G** $y = -x$	**H** $y = -x - 2$
I $y = 0{,}5x + 1$	**J** $y = 0{,}5x - 2$	**K** $y = -0{,}5x + 3$	**L** $y = -0{,}5x$
M $y = 2$	**N** $y = -1$	**O** $y = 3x$	**P** $y = -2x + 1$

Snakk om etterpå: Hvilke spørsmål delte grafene i to like store grupper? Hvorfor er det lurt?

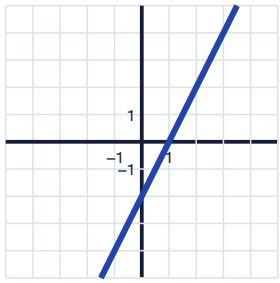
Hvem er grafen? spillebrett

Kryss ut grafene som ikke kan være motstanderens graf.

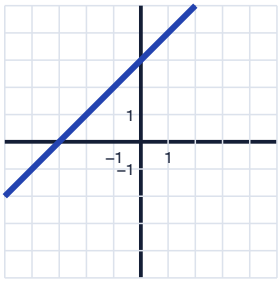
A



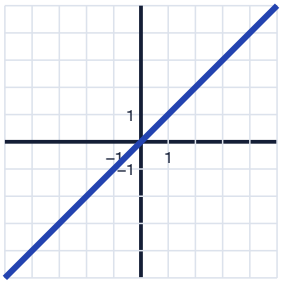
B



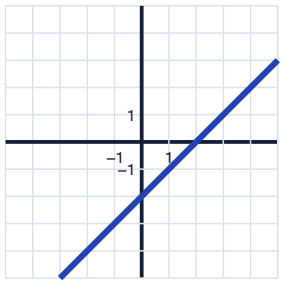
C



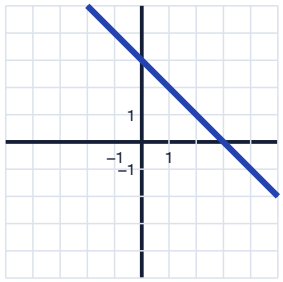
D



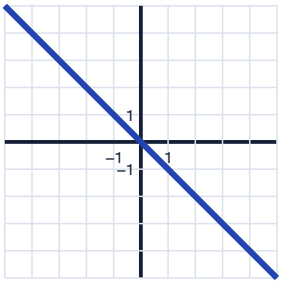
E



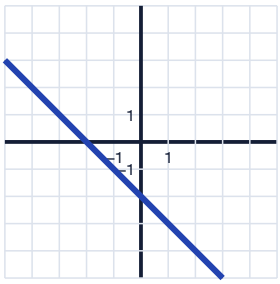
F



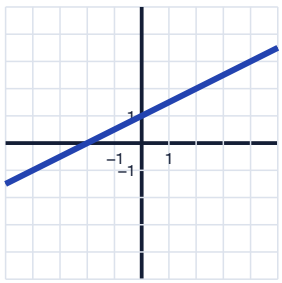
G



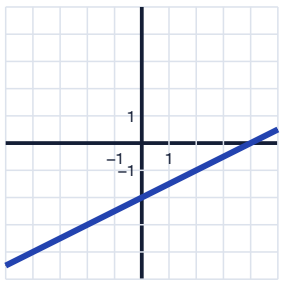
H



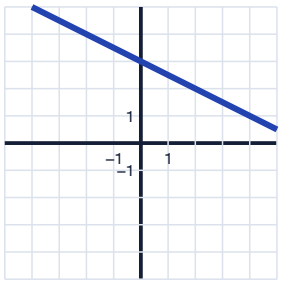
I



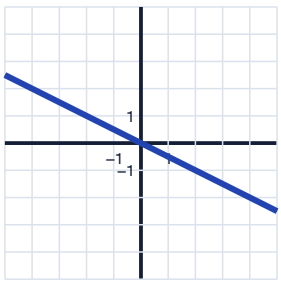
J



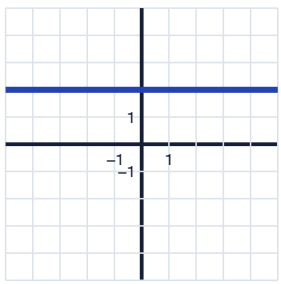
K



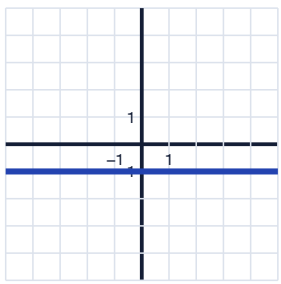
L



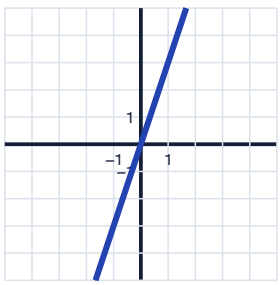
M



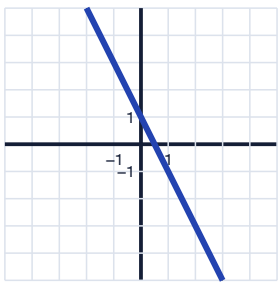
N



O



P

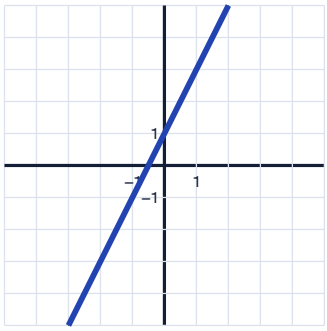


Hvem er grafen?

✂ Klipp langs de stiplede linjene

HEMMELIG GRAF

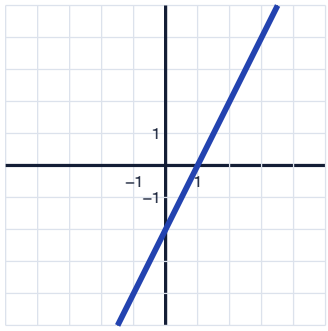
A



y = 2x + 1

HEMMELIG GRAF

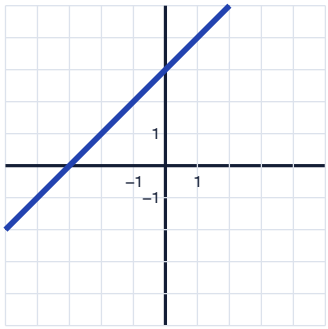
B



y = 2x − 2

HEMMELIG GRAF

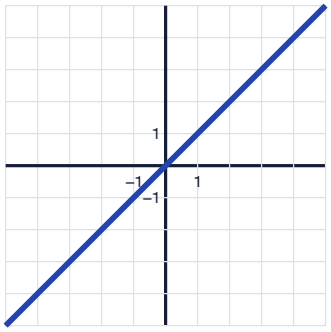
C



y = x + 3

HEMMELIG GRAF

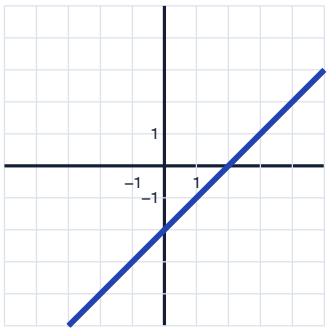
D



y = x

HEMMELIG GRAF

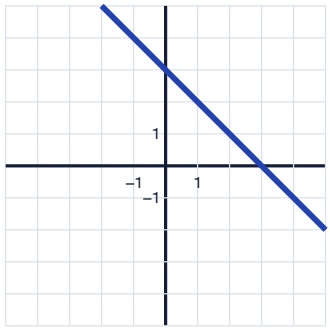
E



y = x − 2

HEMMELIG GRAF

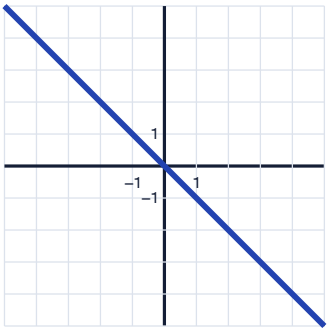
F



y = −x + 3

HEMMELIG GRAF

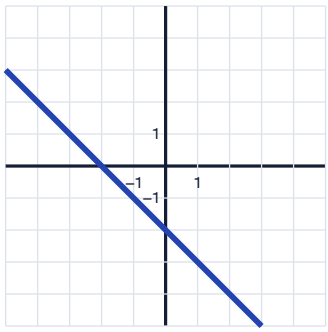
G



y = −x

HEMMELIG GRAF

H

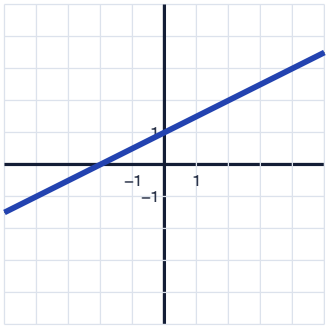


y = −x − 2

Hvem er grafen?

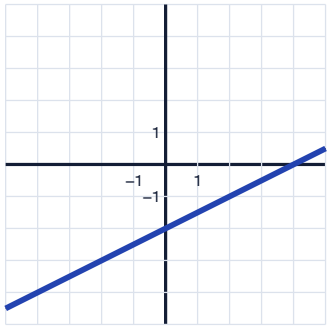
✂ Klipp langs de stiplede linjene

HEMMELIG GRAF **I**



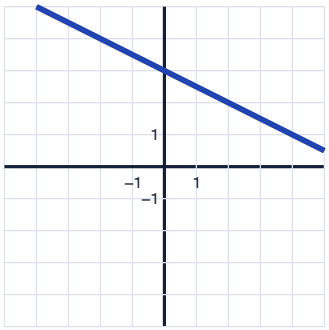
y = 0,5x + 1

HEMMELIG GRAF **J**



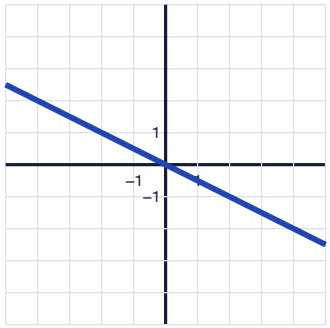
y = 0,5x − 2

HEMMELIG GRAF **K**



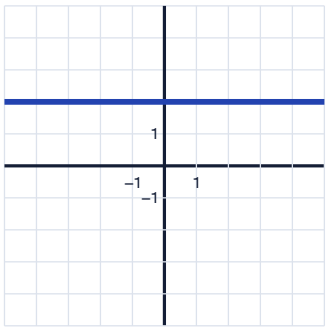
y = −0,5x + 3

HEMMELIG GRAF **L**



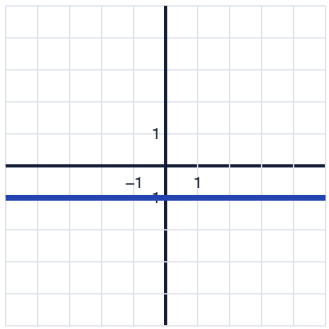
y = −0,5x

HEMMELIG GRAF **M**



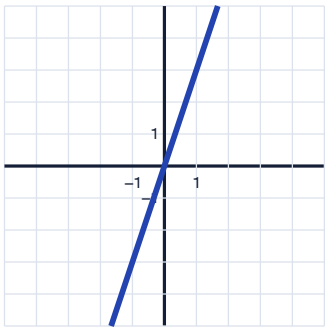
y = 2

HEMMELIG GRAF **N**



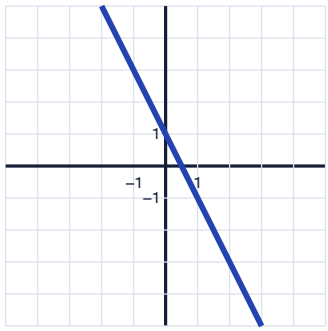
y = −1

HEMMELIG GRAF **O**



y = 3x

HEMMELIG GRAF **P**



y = −2x + 1